Neutrino-Nucleon Structure Functions in Duality and DIS Regions

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Abstract

Neutrino-nucleon charged-current deep inelastic scattering (DIS) in the transition region ($Q^2 \sim$ 1–4 GeV^2) is important for neutrino oscillation experiments such as NOvA, T2K, DUNE, and Hyper-Kamiokande, where perturbative QCD breaks down and non-perturbative hadronic effects become significant. Motivated by Bloom–Gilman quark–hadron duality, we construct a hybrid description of F_2 and xF_3 across this region. We combine the Regge-inspired CKMT parametrization at low Q^2 with the AMU perturbative QCD framework (NLO + target mass corrections + higher-twist effects) at higher Q^2. The models are matched at changeover scale $Q_0^2 = 2$ GeV^2 using two established transition prescriptions: a piecewise linear splice and a smooth sigmoid weighting function. Linear splicing of unscaled CKMT with AMU generates unphysical kink artefacts in xF_3 at low x due to curvature and normalization differences between the models. The smooth sigmoid weighting ($\Delta = 0.25$ GeV^2, AMU with LO+NLO+TMC+HT) restores derivative continuity across the matching boundary; however, a non-monotonic dip persists due to the normalization offset between unscaled CKMT and AMU, which is further deepened when higher-twist corrections are included. The hybrid framework highlights the necessity of smooth matching and proper normalization across the resonance-to-DIS interface.

Keywords: neutrino deep inelastic scattering; weak structure functions; CKMT parametrization; low-Q^2 modelling; quark–hadron duality

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1 Introduction

Neutrino deep inelastic scattering (DIS) spans quasi-elastic, resonance, and perturbative regimes. Its few-GeV transition region ($Q^2 \sim$ 1–4 GeV^2) is important for oscillation experiments including NOvA, T2K, DUNE, and Hyper-Kamiokande, yet no single model describes it consistently [1].

Bloom–Gilman quark–hadron duality [2], and its extension to neutrino-induced weak processes [3], motivates matching low-Q^2 empirical parametrizations to perturbative QCD (pQCD). Resonance structure, plotted against the modified scaling variable $\omega' = (2M\nu + M^2)/Q^2$, averages around the DIS scaling curve. Generators such as GENIE [4] and NuWro [5] use the Bodek–Yang prescription [6] to extend pQCD into this region; it relies on GRV98 LO PDFs and treats target-mass and higher-twist effects implicitly. Here we compare linear and smooth-sigmoid matching of the CKMT [7] and AMU [8] (NLO+TMC+HT) models for free-nucleon F_2 and xF_3; extension to nuclear targets is ongoing.

2 Theoretical Framework

2.1 Charged-Current DIS Cross Section

The charged-current hadronic tensor is described by five dimensionless structure functions, $F_i(x,Q^2)$ $(i = 1,\ldots,5)$. For charged-current neutrino scattering $\nu_\mu(k) + N(p) \to \mu^-(k') + X(p')$, the double-differential cross section for a nucleon at rest, including the outgoing-lepton mass, is [9]

$$\begin{aligned}\frac{d^2\sigma^{\nu,\bar{\nu}}}{dx\,dy} = \frac{G_F^2 M_N E_\nu}{\pi}\left(\frac{M_W^2}{Q^2+M_W^2}\right)^2 \Bigg\{ &\left(y^2 x + \frac{m_\ell^2 y}{2M_N E_\nu}\right) F_1 \\ &+ \left[1 - \frac{m_\ell^2}{4E_\nu^2} - \left(1 + \frac{M_N x}{2E_\nu}\right) y\right] F_2 \\ &\pm xy\left(1 - \frac{y}{2} - \frac{m_\ell^2 y}{4M_N E_\nu}\right) F_3 \\ &+ \frac{m_\ell^2(m_\ell^2+Q^2)}{4E_\nu^2 M_N^2 x} F_4 - \frac{m_\ell^2}{E_\nu M_N} F_5 \Bigg\},\end{aligned} \tag{1}$$

where G_F is the Fermi coupling constant, m_ℓ is the mass of lepton, E_ν is the incident neutrino (or antineutrino) energy, and M_N is the nucleon mass. The Bjorken scaling variable is defined as $x = Q^2/(2M_N\nu)$, where ν is the energy trans-

fer, q is the four-momentum transfer, and $Q^2 = -q^2$. The inelasticity variable is given by $y = \nu/E_\nu$. The structure functions F_4 and F_5 are neglected because their contributions are suppressed by the charged-lepton mass. At leading order, F_1 and F_2 are related through the Callan–Gross relation $F_2(x,Q^2) = 2xF_1(x,Q^2)$, leaving F_2 and F_3 as the two independent structure functions. The contribution of F_3 enters with opposite signs for neutrino and antineutrino scattering.

The nucleon structure functions are calculated using parton distribution functions (PDFs) within the framework of perturbative QCD. In the present work, the CT10nlo PDF set [10] is employed to describe the quark and antiquark distributions in the nucleon. Next-to-leading-order (NLO) QCD corrections are incorporated using the coefficient-function calculations of Vermaseren, Vogt, and Moch, and van Neerven and Vogt [11–13]. Furthermore, target mass corrections (TMCs) [14] are included to account for finite nucleon-mass effects, which are important at high x and moderate Q^2. At low Q^2 and high x, we also include phenomenological higher-twist (HT) contributions, which account for power-suppressed effects beyond the leading-twist description. Their implementation in the AMU model is described below.

2.2 AMU (High-Q^2) Model

Related AMU calculations of weak nucleon and nuclear structure functions with perturbative and nonperturbative QCD corrections are presented in Ref. [8]. Here, the structure functions are computed with CT10nlo PDFs [10], NLO coefficient functions, target-mass corrections, and higher-twist contributions as described below.

The partons inside the nucleon interact through gluon exchange, as described by QCD. Including gluon emission introduces a dependence on Q^2 in the nucleon structure functions and thus violates Bjorken scaling. At finite Q^2, the strong coupling can make NLO contributions significant, with higher perturbative orders also contributing. Following Vermaseren et al. [11], Moch et al. [12], and van Neerven and Vogt [13], the structure functions are expressed in terms of the PDFs and the quark- and gluon-level coefficient functions:

$$\begin{aligned} x^{-1}F_{2,L}(x,Q^2) &= \sum_{f=q,g}^{\infty} C_{2,L}(\alpha_s)\otimes f_2, \\ F_3(x,Q^2) &= \sum_{f=q,g}^{\infty} C_3(\alpha_s)\otimes f_3, \end{aligned} \tag{2}$$

where C_2 and C_3 are the coefficient functions for quarks and gluons, $\otimes$ denotes the Mellin convolution (which becomes multiplication in Mellin-N space), and f_2 and f_3 represent the quark and gluon distributions, respectively.

For the target-mass corrections, we use the Nachtmann variable $\xi = 2x/(1+\gamma)$, with $\gamma = \sqrt{1+4x^2\mu}$ and $\mu = M_N^2/Q^2$. The expressions used for the three structure functions are [14]:

$$\begin{aligned} F_1^{\mathrm{TMC}}(x,Q^2) &\simeq \frac{x}{\xi\gamma}F_1(\xi,Q^2)\left[1+\frac{2\mu x\xi}{\gamma}(1-\xi)^2\right], \\ F_2^{\mathrm{TMC}}(x,Q^2) &\simeq \frac{x^2}{\xi^2\gamma^3}F_2(\xi,Q^2)\left[1+\frac{6\mu x\xi}{\gamma}(1-\xi)^2\right], \\ F_3^{\mathrm{TMC}}(x,Q^2) &\simeq \frac{x}{\xi\gamma^2}F_3(\xi,Q^2)\left[1-\frac{\mu x\xi}{\gamma}(1-\xi)\ln\xi\right]. \end{aligned} \tag{3}$$

Here the structure functions on the right-hand sides are evaluated at the same Q^2 and at the rescaled argument ξ.

The leading-twist structure function, including TMCs, is supplemented by the phenomenological higher-twist term:

$$F_2(x,Q^2) = F_2^{\mathrm{LT+TMC}}(x,Q^2)\left[1+\frac{D_2(x,Q^2)}{Q^2}\right], \tag{4}$$

where D_2 parameterizes multiparton correlations and has dimensions of mass squared [15, 16]. The $1/Q^2$ dependence makes this correction most relevant at low Q^2.

We use the AMU model for $Q^2 \gtrsim 2$ GeV^2; at lower Q^2, the perturbative description is not applicable.

2.3 CKMT Parametrization

The CKMT model [7] is a Regge-inspired empirical parametrization originally fitted to electromagnetic F_2 data, which is adapted to charged-current neutrino scattering for F_2^{CC} and xF_3 [17]. The electromagnetic structure function is written as a sum of sea and valence terms:

$$\begin{aligned} F_2^{\mathrm{CKMT}}(x,Q^2) &= F_2^{\mathrm{sea}}(x,Q^2) + F_2^{\mathrm{val}}(x,Q^2) \\ &= Ax^{-\Delta(Q^2)}(1-x)^{n(Q^2)+4} \\ &\quad\times\left(\frac{Q^2}{Q^2+a}\right)^{1+\Delta(Q^2)} \\ &\quad + Bx^{1-\alpha_R}(1-x)^{n(Q^2)} \\ &\quad\times\left(\frac{Q^2}{Q^2+b}\right)^{\alpha_R}[1+f(1-x)], \end{aligned} \tag{5}$$

where

$$n(Q^2) = \frac{3}{2}\left(1+\frac{Q^2}{Q^2+c}\right), \qquad \Delta(Q^2) = \Delta_0\left(1+\frac{2Q^2}{Q^2+d}\right). \tag{6}$$

For F_2^{CC} and xF_3, the same functional form is used with the channel-dependent values of A, B, and f in Table 1; the other parameters are common.

2.4 Hybrid Matching: Linear Splice

To bridge the low-Q^2 CKMT regime and high-Q^2 pQCD, the two models are matched across a crossover region around $Q_0^2 = 2.0$ GeV^2. Here, "unscaled CKMT" denotes the original parametrization $F^{\mathrm{CKMT}}(x,Q^2)$. For comparison, "scaled CKMT" denotes a multiplicatively normalized version. For

$\Delta_0 = 0.07684$	$\alpha_R = 0.4150$	$a = 0.2631$ GeV^2
$b = 0.6452$ GeV^2	$c = 3.5489$ GeV^2	$d = 1.1170$ GeV^2

Process	A	B	f
EM F_2	0.1502	1.2064	0.15
νN F_2	0.5967	2.7145	0.5962
νN xF_3	9.3955×10^{-3}	2.4677	0.5962
$\bar{\nu} N$ xF_3	9.3955×10^{-3}	-2.4677	0.5962

Table 1: CKMT parameters used in the electromagnetic and charged-current structure functions [17]. B and f are fixed by valence constraints at $Q^2 = 2$ GeV^2.

each charged-current structure function $F_{i,\mathrm{CC}}$ ($i = 2, 3$), we use the normalization prescription

$$F_{i,\mathrm{CC}}^{\mathrm{scaled}}(x,Q^2) = F_{i,\mathrm{CC}}^{\mathrm{CKMT}}(x,Q^2)\,\frac{F_{i,\mathrm{CC}}^{\mathrm{NLO+TMC}}(x,Q_0^2)}{F_{i,\mathrm{CC}}^{\mathrm{CKMT}}(x,Q_0^2)}, \tag{7}$$

where the ratio is evaluated separately at each x and for each structure function, using the NLO calculation with target-mass corrections and without higher-twist corrections in the numerator. It is independent of Q^2, so the scaled CKMT prediction agrees with NLO+TMC at $Q^2 = Q_0^2$. The hybrid prescriptions below use unscaled CKMT; the scaled version is shown only for comparison. A piecewise linear interpolation splice across $[1.5, 2.5]$ GeV^2 is defined by:

$$F^{\mathrm{hyb}} = \begin{cases} F^{\mathrm{CKMT}} & Q^2 \le 1.5 \text{ GeV}^2, \\ c\,F^{\mathrm{AMU}} + (1-c)\,F^{\mathrm{CKMT}} & 1.5 < Q^2 < 2.5 \text{ GeV}^2, \\ F^{\mathrm{AMU}} & Q^2 \ge 2.5 \text{ GeV}^2, \end{cases} \tag{8}$$

with the linear weight $c(Q^2) = (Q^2 - 1.5)/(2.5 - 1.5)$.

2.5 Hybrid Matching: Smooth Sigmoid

Although the linear splice provides value continuity, differences in model curvature can induce derivative discontinuities and kink artefacts. To ensure a continuously differentiable transition, we replace the linear weight with a smooth sigmoid function, yielding the hybrid structure function:

$$F^{\mathrm{hyb}} = w(Q^2)\,F^{\mathrm{CKMT}} + \big[1 - w(Q^2)\big]\,F^{\mathrm{AMU}}, \tag{9}$$

with $w(Q^2) = \frac{1}{2}\big[1 - \tanh\big((Q^2 - Q_0^2)/\Delta\big)\big]$. We use a transition width $\Delta = 0.25$ GeV^2 centred at $Q_0^2 = 2.0$ GeV^2.

3 Results and Discussion

3.1 Hybrid Structure Functions: Linear Splice vs. Smooth Sigmoid

Figure 1 compares the original (unscaled) CKMT, scaled CKMT at $Q_0^2 = 2$ GeV^2, AMU, and the linear splice for F_2. For xF_3, because the linear splice directly joins the unscaled CKMT with AMU, the curvature and normalization mismatch generates a prominent downward kink and slope discontinuity at low x, visible in Fig. 2. Fig. 3 shows the result of replacing the linear weight with a smooth sigmoid ($\Delta = 0.25$ GeV^2). While the sigmoid restores derivative continuity and removes the sharp angular vertices, an unphysical non-monotonic dip remains visible across the transition window $[1.5, 2.5]$ GeV^2 because unscaled CKMT sits systematically above AMU at $Q^2 \sim 2$ GeV^2. Furthermore, HT corrections suppress AMU by 5–15% (purple dash-dotted vs. orange dashed), widening this normalization gap and deepening the dip in the full hybrid curve (green solid).

4 Conclusions

We have presented a hybrid framework for the charged-current weak structure functions F_2 and xF_3 that bridges the non-perturbative CKMT description at low Q^2 and the AMU (NLO+TMC+HT) pQCD calculation at high Q^2, with a changeover scale $Q_0^2 = 2.0$ GeV^2. Our results show that linear splicing can produce visible kinks in xF_3, particularly at low x. Smooth sigmoid weighting with $\Delta = 0.25$ GeV^2 removes the sharp slope discontinuity, but a non-monotonic dip remains because of the normalization difference between unscaled CKMT and AMU, and is accentuated by higher-twist corrections. These results emphasize that a smooth transition alone is not sufficient: the normalization of the two descriptions must also be addressed. Future work will validate the framework against CCFR/NuTeV structure-function data, extend it to additional nuclear targets, and include PCAC/axial contributions.

Conflict of Interest

The authors declare that they have no conflict of interest.

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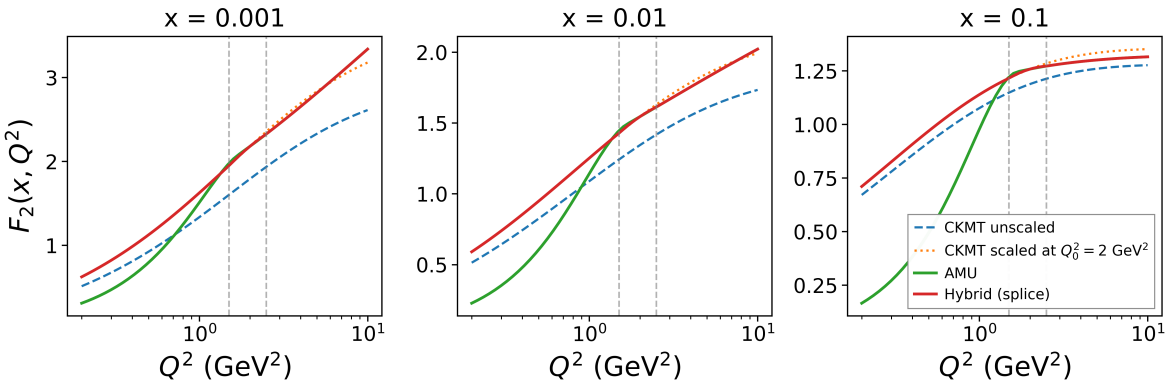


Figure 1: $F_2(x, Q^2)$ vs. Q^2 at $x = 0.001, 0.01, 0.1$ for unscaled CKMT (blue dashed), scaled CKMT at $Q_0^2 = 2$ GeV^2 (orange dotted), AMU (green solid), and linear splice (red solid). Vertical dashed lines mark the splice window $[1.5, 2.5]$ GeV^2.

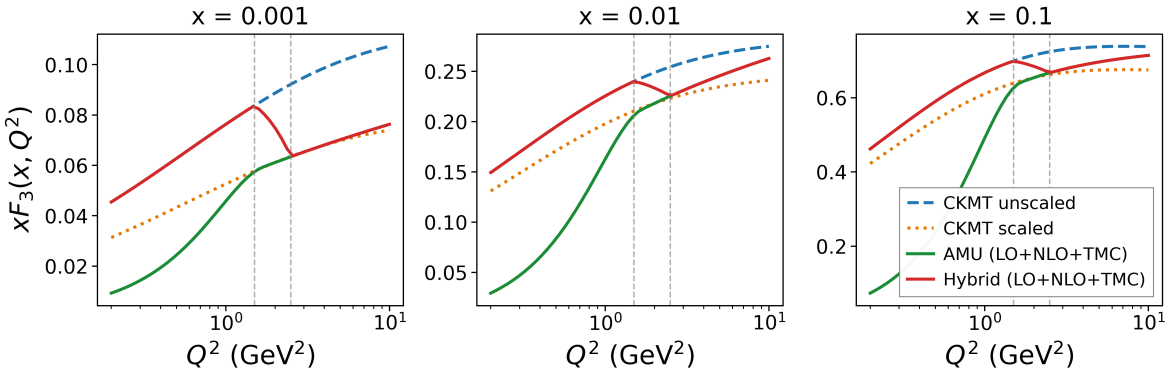


Figure 2: $xF_3(x, Q^2)$ vs. Q^2 for the linear splice: unscaled CKMT (blue dashed), CKMT scaled at $Q_0^2 = 2$ GeV^2 (orange dotted), AMU (LO+NLO+TMC; green solid), and the hybrid (red solid). The kink and slope discontinuity are most prominent at low x.

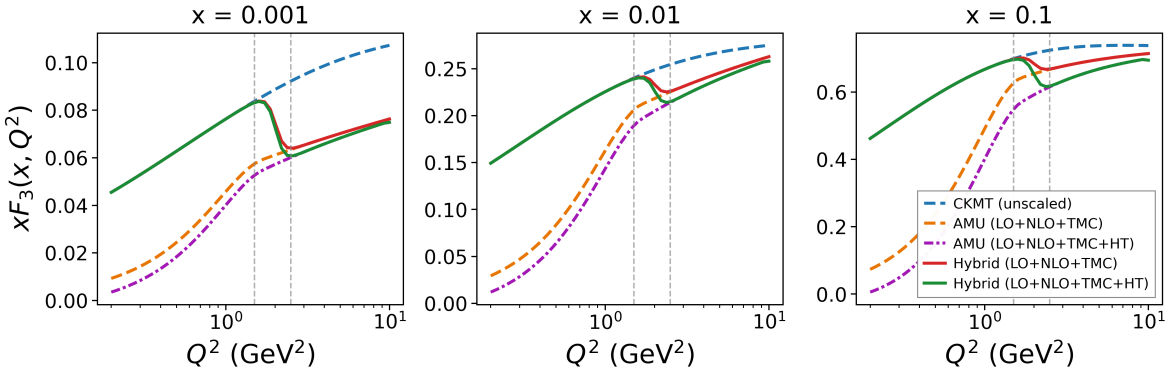


Figure 3: $xF_3(x, Q^2)$ vs. Q^2 with smooth sigmoid matching ($\Delta = 0.25$ GeV^2). Derivative continuity is restored, but the normalization offset leaves a non-monotonic dip; HT corrections deepen it slightly.

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